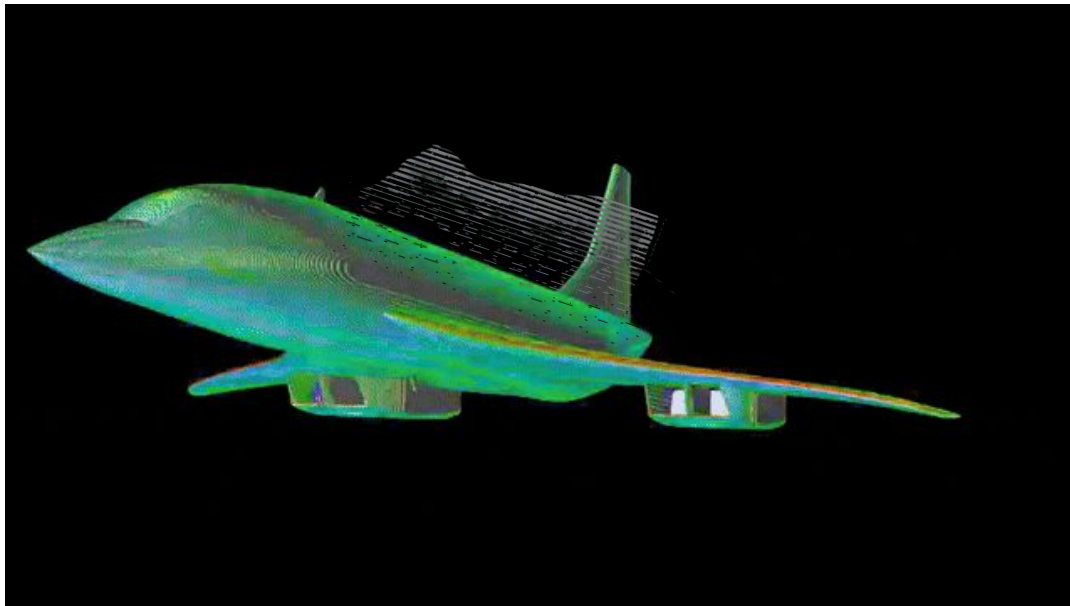


A Cognitive Modelling Framework for Simulating Human-Autonomy Teaming in the Flight Deck

Benjamin R. Berton, Philippe Doyon-Poulin, Polytechnique Montréal

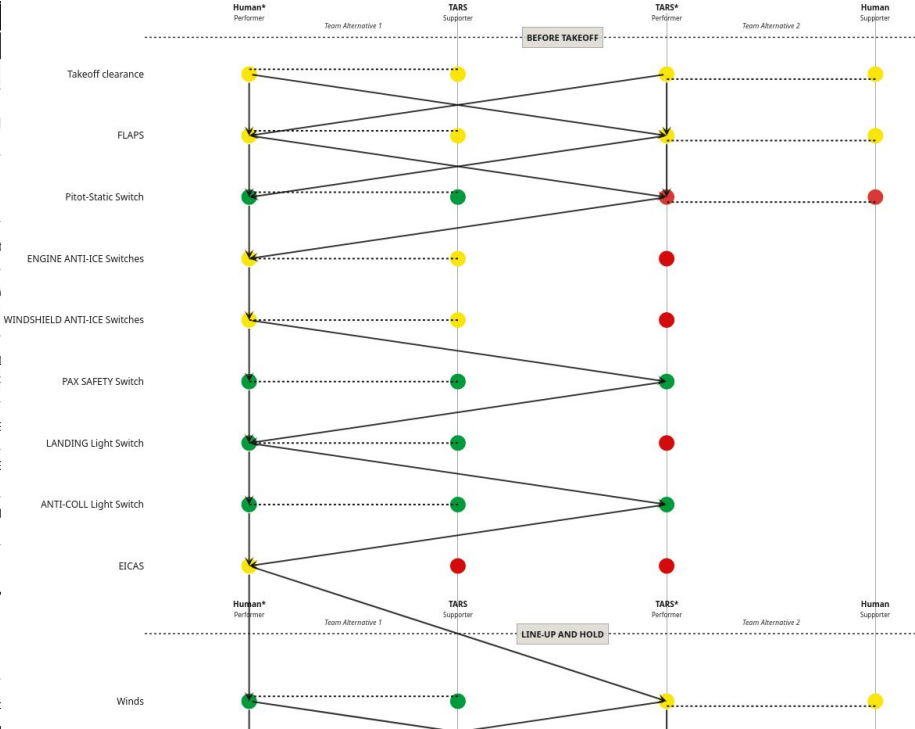
Corresponding author: benjaminberton64@gmail.com

My personal motivation



Motivation

ACTIVITY DECOMPOSITION					
Row	Procedure	Type	Category	Task	
1	BEFORE TAKEOFF	Checklist	Switch / Lever	Takeoff c	
2		Checklist	Switch / Lever	FLAPS	
3		Checklist	Switch / Lever	Pitot-Stat	
4		Checklist	Switch / Lever	ENGINE / Switches	
5		Checklist	Switch / Lever	WINDSHI ICE Switc	
6		Checklist	Switch / Lever	PAX SAFE	
7		Checklist	Switch / Lever	LANDING Switch	
8		Checklist	Switch / Lever	ANTI-COI Switch	
9		Checklist	Parameter Check	EICAS	
10	LINE-UP AND HOLD	SOP	Environment Check	Winds	
11		SOP	Autopilot Mgmt	Select Alt	



TEAMING REQUIREMENTS	
Predictability	Directability
	Pilot can command TARS to recheck an item.
Pilot anticipates TARS's flap setting callout before takeoff.	Pilot can confirm or override TARS's flap setting.
TARS alerts pilot if anti-ice status deviates from requirement.	
Pilot expects TARS to suggest if it is suggested based on time of day.	
Pilot expects TARS to alert if ANTI-COLL light switch is off.	
	Both can perform in parallel and crosscheck

Motivation

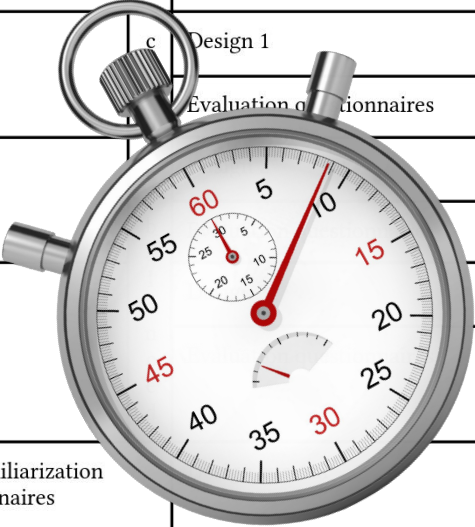
To design effective teaming, we must test, fail, learn, refine. But in aviation, empirical evaluation has a cost.



Motivation

Recruiting qualified pilots is difficult and time consuming. Experimental sessions are long (≈ 4 hours per participant for ≈ 4 condition), which limits the number of conditions/configurations that can be evaluated per study.

Briefing (30min) →	Training (1h) ↓	Experimental block (2h) ↓	Debriefing (45min)
	Design 1	Design 1	c o u n t e r b a l a n c e d
		Evaluation questionnaires	
	Design 2		
	Design 3		
	Post-familiarization questionnaires		
		Evaluation questionnaires	

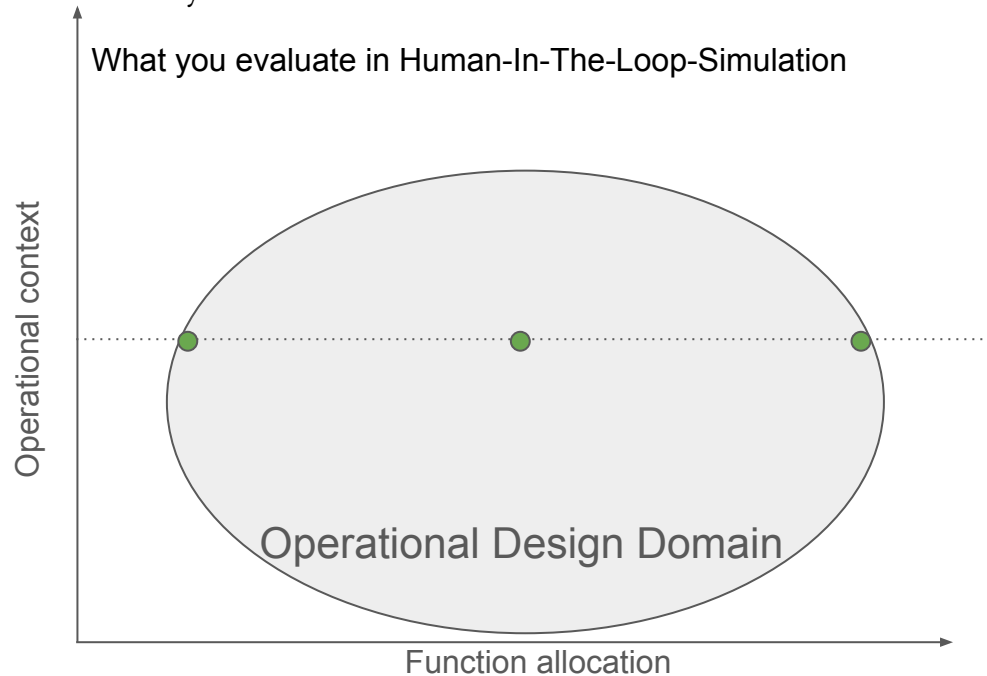


Motivation

To avoid confounds, traditional Human Factors studies “freeze” the operational context.

Standardize: aircraft state/configuration, failure set, timing, weather, traffic/ATC/navigation.

Limited coverage of real-world variability



1

FRAMEWORK OVERVIEW



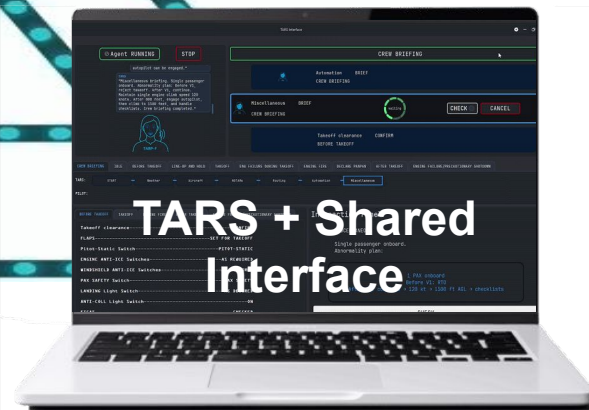
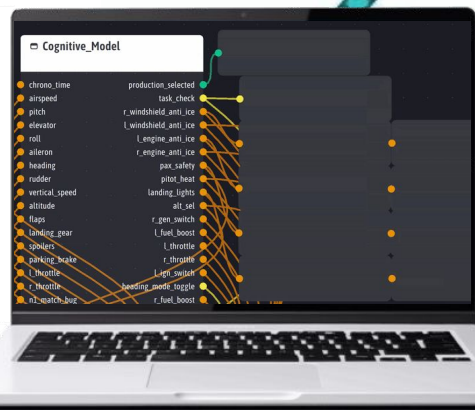
flight commands

cockpit instruments
reading

avionics commands
avionics data

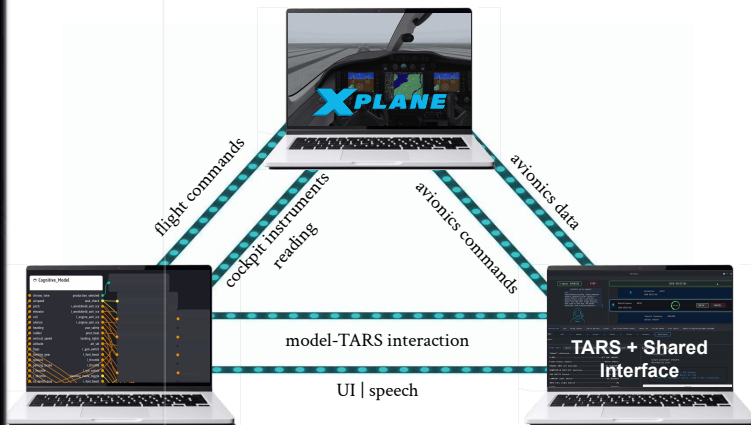
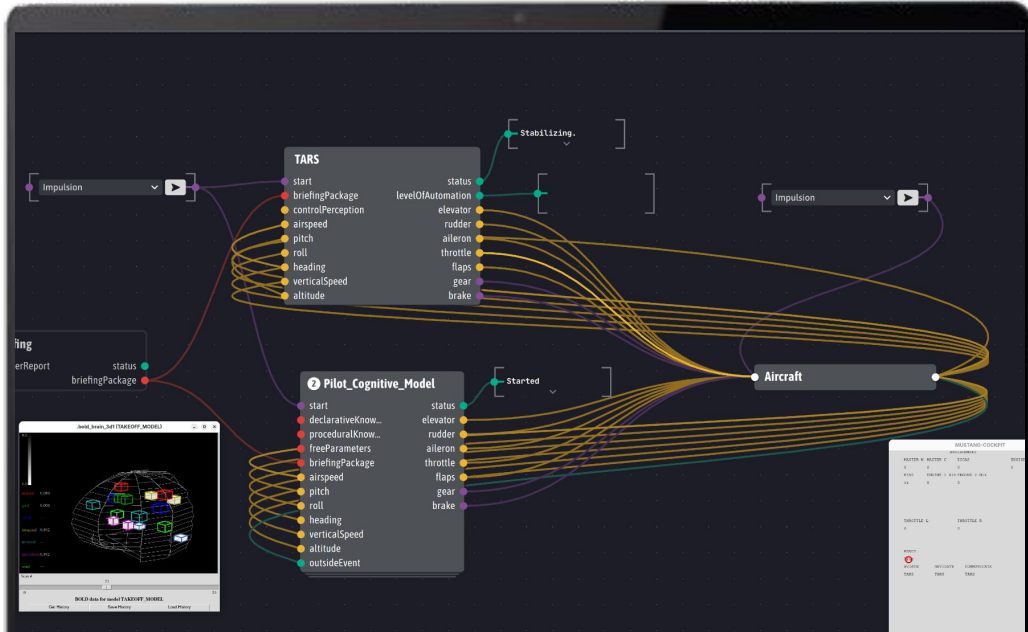
model-TARS interaction

UI | speech



1

FRAMEWORK OVERVIEW

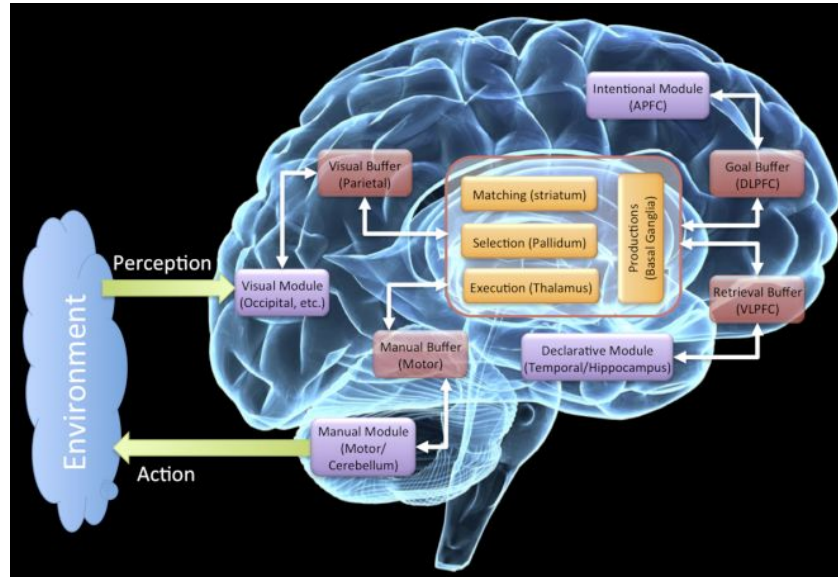
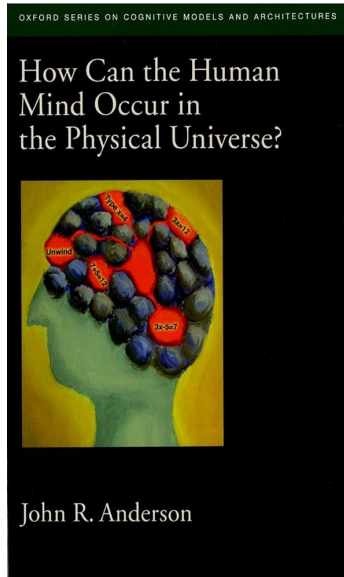


COGNITIVE MODEL

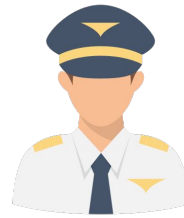


What is it?

ACT-R (Adaptive Control of Thought–Rational) is a cognitive architecture, a theory about how the human mind functions at the cognitive level.

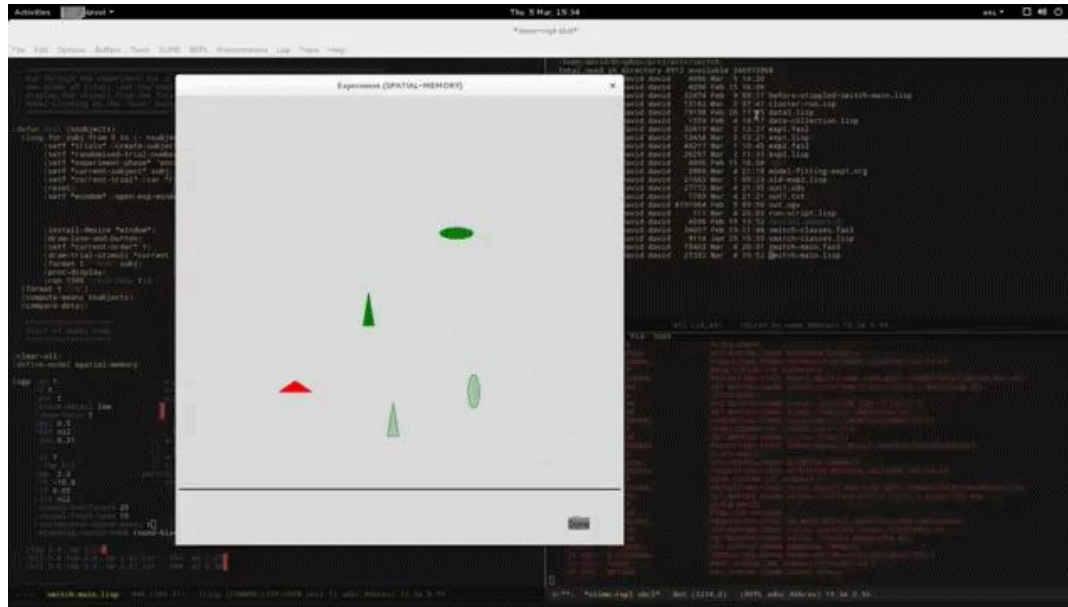


COGNITIVE MODEL



What is it?

This theory has an implementation in computer code, that can be used to predict human performance on various cognitive tasks.



COGNITIVE MODEL

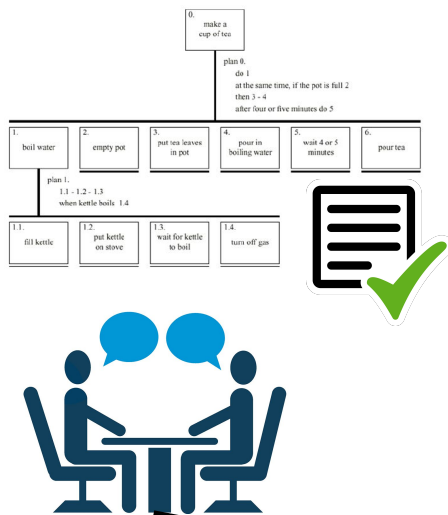


How did we build it?

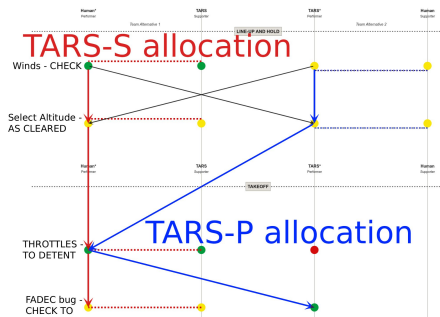
Operational documentation and training manuals



Validation with 3 expert pilots



Interdependence Analysis output



Teaming requirements:

- Observability
- Predictability
- Directability

declarative +
procedural
knowledge

ACT-R

COGNITIVE MODEL



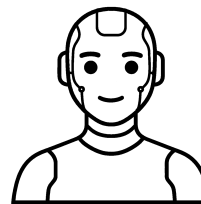
How did we build it?

Human-automation interaction signatures (from prior work) were encoded in the cognitive model.

1. **Delegation $\neq$ less cognition:** shifts from execution to monitoring/verification [29,6,23,21]
 - Model: stochastic supervision / cross-checking of delegated actions
2. **Verification varies across individuals:** [21]
 - Model: captured via an initial verification propensity parameter
3. **Verification rate is dynamic:** number of crosscheck decreases after experiencing reliable automation, conversely increases after experiencing unreliable automation. [3,19,23,28]
 - Model: reward/punishment updates the verification propensity parameter.
4. **Support is integrated on the fly:** operators proceed and opportunistically integrate support when it arrives.
 - Model: continue independently; integrate support when available [18,38]
5. **Attention allocation:** visual attention driven by SEEV-based probabilities (pre-tuned from prior estimates) [32,37,31]

Outcome: reproduces known cognitive signatures such as adaptive reliance, complacency, distrust after failures.

TARS, THE FLIGHT DECK AGENT



Takeoff Autonomous Resilience System (TARS) supports *or* performs flight-deck tasks during takeoff.

Implementation: **reactive finite-state machine** (Python).

Behavior is parameterized via **task-allocation files** (per task: *support vs. perform vs. nothing; immediate vs. delayed*).

Capabilities

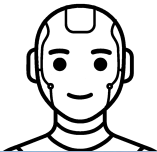
- Reads aircraft state
- Executes checklist actions and monitors switch states
- Interfaces with selected avionics
- Communicates via voice and text

Shared interface (TARS UI - PyQt)

- Implements teaming requirements from Interdependence Analysis
- Displays: task state, agent state, allocation, upcoming actions, task-specific cues, and intervention controls
- Deployed on a 17-inch touchscreen; the cognitive model receives a text-based UI description

3

TARS, THE FLIGHT DECK AGENT

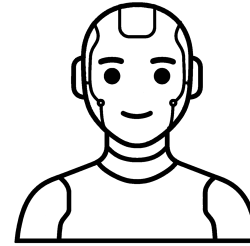


FRAMEWORK PARAMETERS



Cognitive model (pilot)

- Architecture parameters: ACT-R (defaults)
- Manual action timing: latencies for cockpit actions
- Initial verification propensity: 0 → always verify, 1 → never verify (default: 0.5)
- Adaptive reliance learning rate: reward weights for correct vs. incorrect agent actions (updates verification behavior)
- Attention allocation: SEEV parameters for AoIs
 - Saliency
 - Expectation
 - Effort
 - Value



Autonomous teammate (TARS)

- Function allocation: nothing vs. support vs. perform (per task)
- Timing: immediate vs. delayed actions (default: 5s)
- Interaction modality: text by default + speech optional
- Reliability: Per task (0 → unreliable, 1 → perfectly reliable)

Flight simulator (X-Plane)

- Weather
- Navigation,
- Traffic,
- Failures,
- ...



FRAMEWORK OUTPUTS

Team-level outcomes

- **Joint Activity Efficiency:** Ratio of time spent actually conducting the task at hand over the complete active task time
- **Coordination overhead:** Time spent strictly for interaction/coordination between the model and the autonomy

Task and scenario performance

- **Time on task**
- **Automation error handling:** detected vs. undetected errors
- **Flight performance:** Aviate / Navigate / Communicate / Manage systems

Team processes and interaction

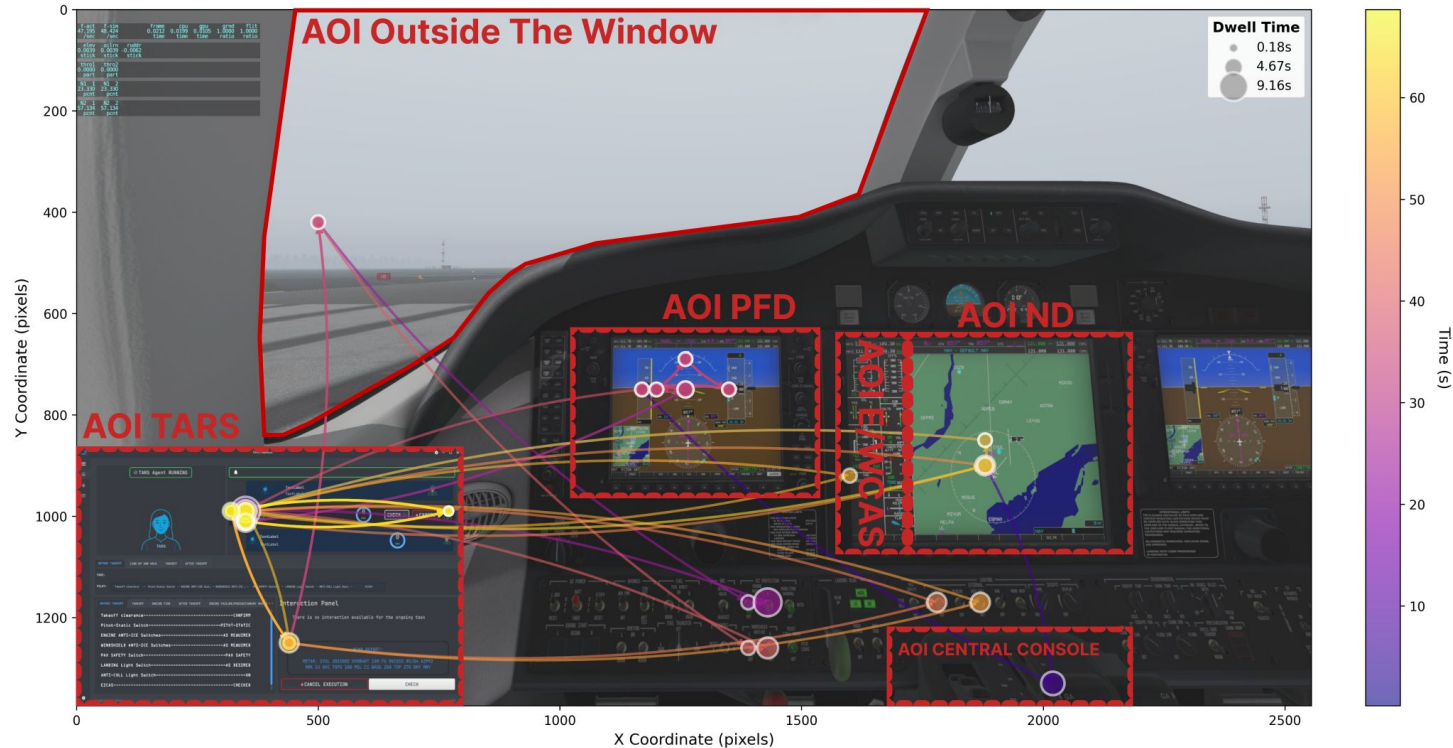
- **Interaction frequency** (# of interventions/communications)
- **Interaction modality:** touch / speech / yoke-button
- **Attention to the shared UI:** dwell time, fixation count, and fixation time on shared interface

Human cognition

- **Workload proxy:** QN-ACTR server utilization
- **Reliance proxy:** production utility (tendency to accept/verify automation actions)
- **Visual attention and gaze pattern**

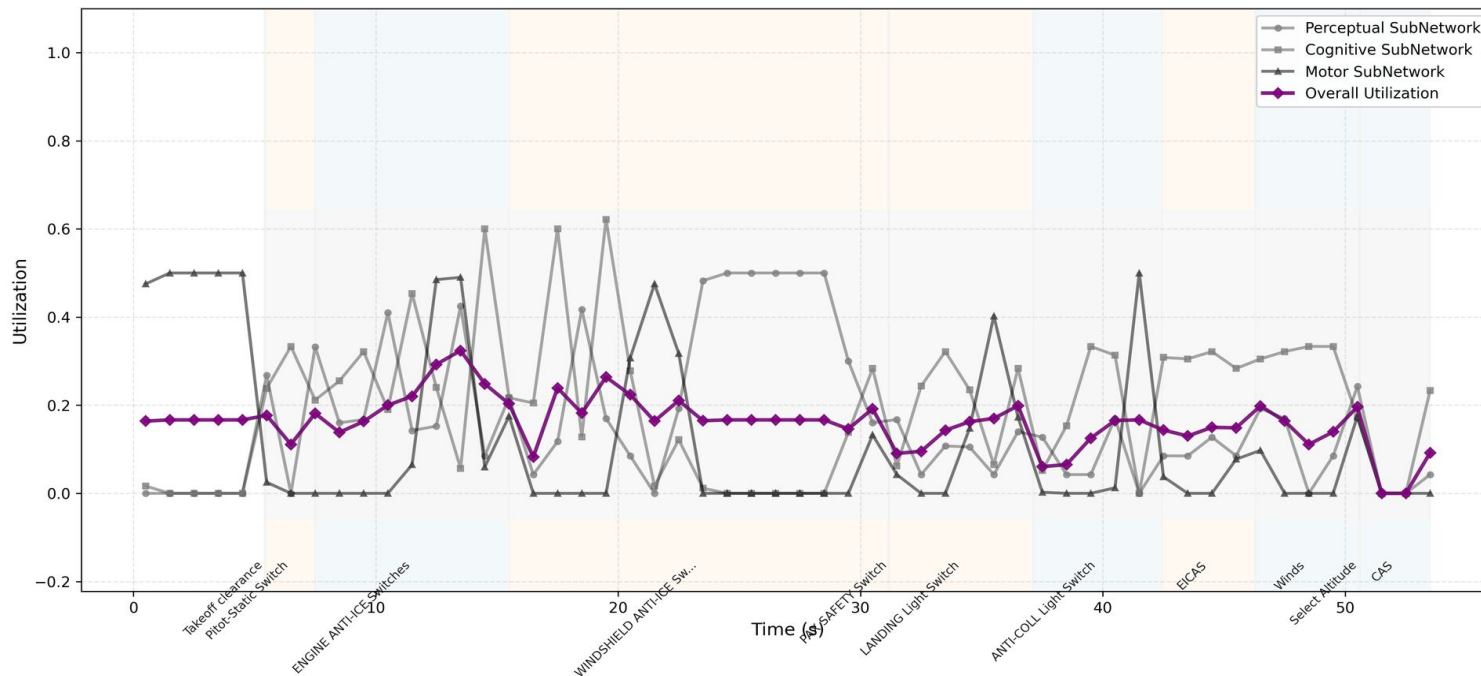
FRAMEWORK OUTPUTS: Human cognition

Model-predicted scanpath during pre-takeoff



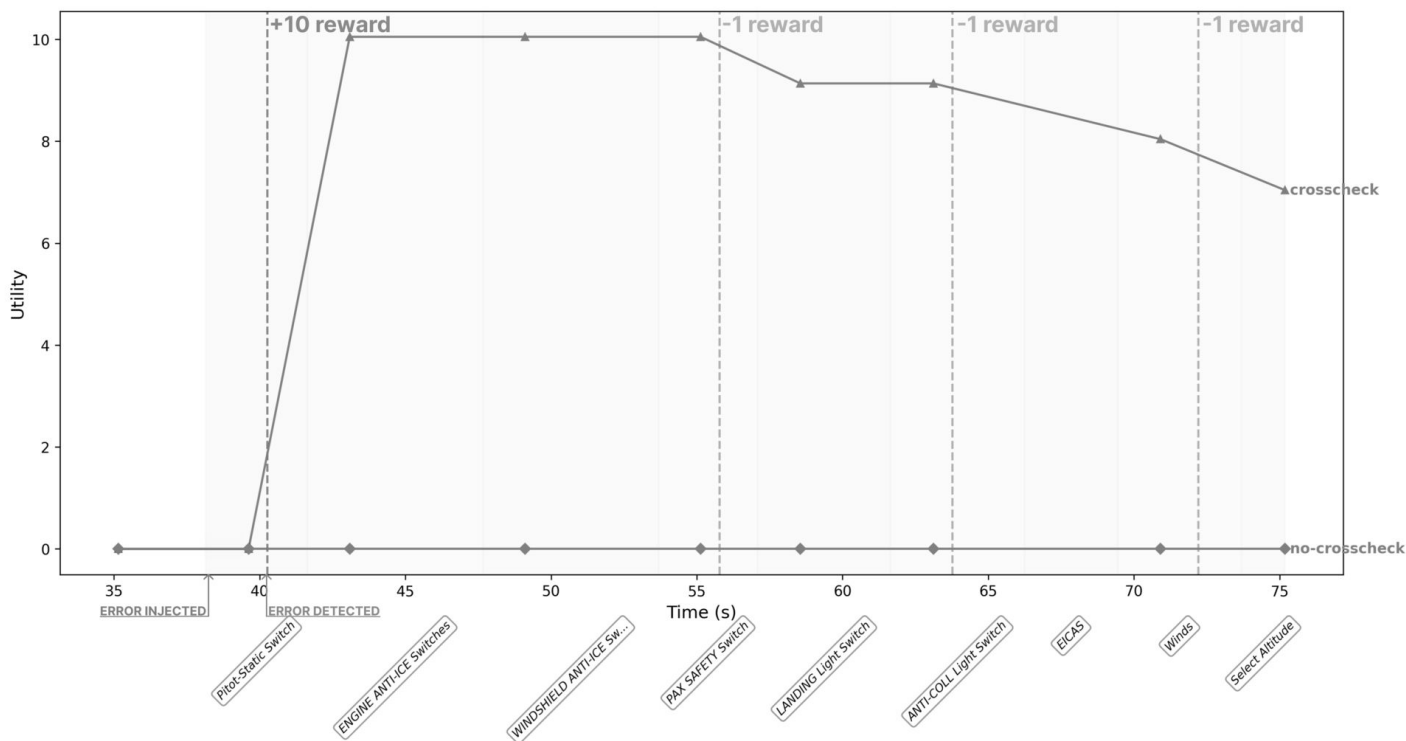
FRAMEWORK OUTPUTS: Human cognition

Model-predicted workload during pre-takeoff



FRAMEWORK OUTPUTS: Human cognition

Propensity to crosscheck delegated action after a TARS unreliable occurrence



6 FUTURE MODEL VALIDATION

Initial simulation run (Dec.)

- 12 runs: 2 allocation conditions $\times$ 2 timing settings (5 s delays vs. none)
- Framework executed without failure, in real time speed
- Outputs were successfully logged; conditions produced distinct performance patterns

Human-in-the-loop study (May)

- N = 18 pilots, Polytechnique flight simulator; same scenario as modeled
- 4 conditions: Support-only, Perform + delays (5 s), Perform without delays, Participant-configured allocation & delays (per task)

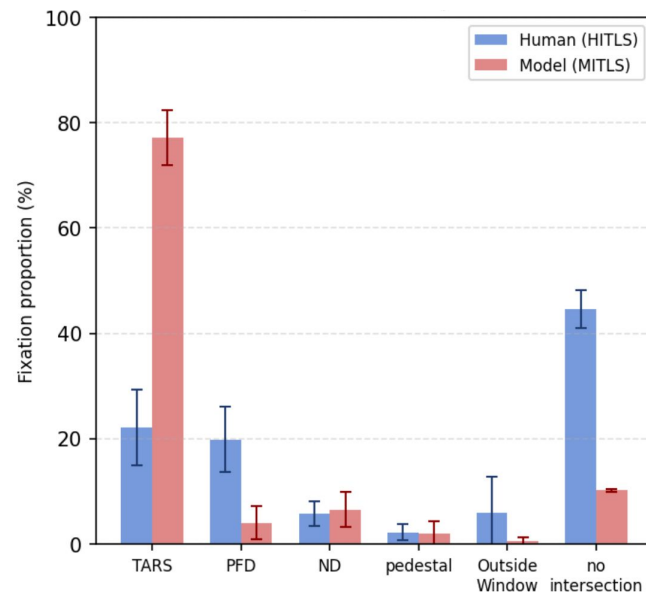
Validation targets (in progress)

- Time on task, Visual attention, Workload

Next:

- cross-check rate, Joint Activity Efficiency, flight performance, etc.

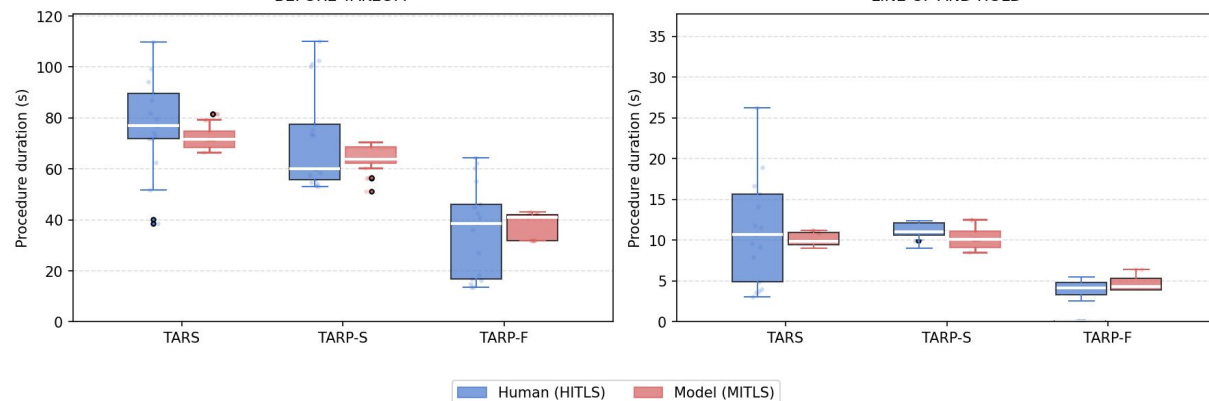
visual attention



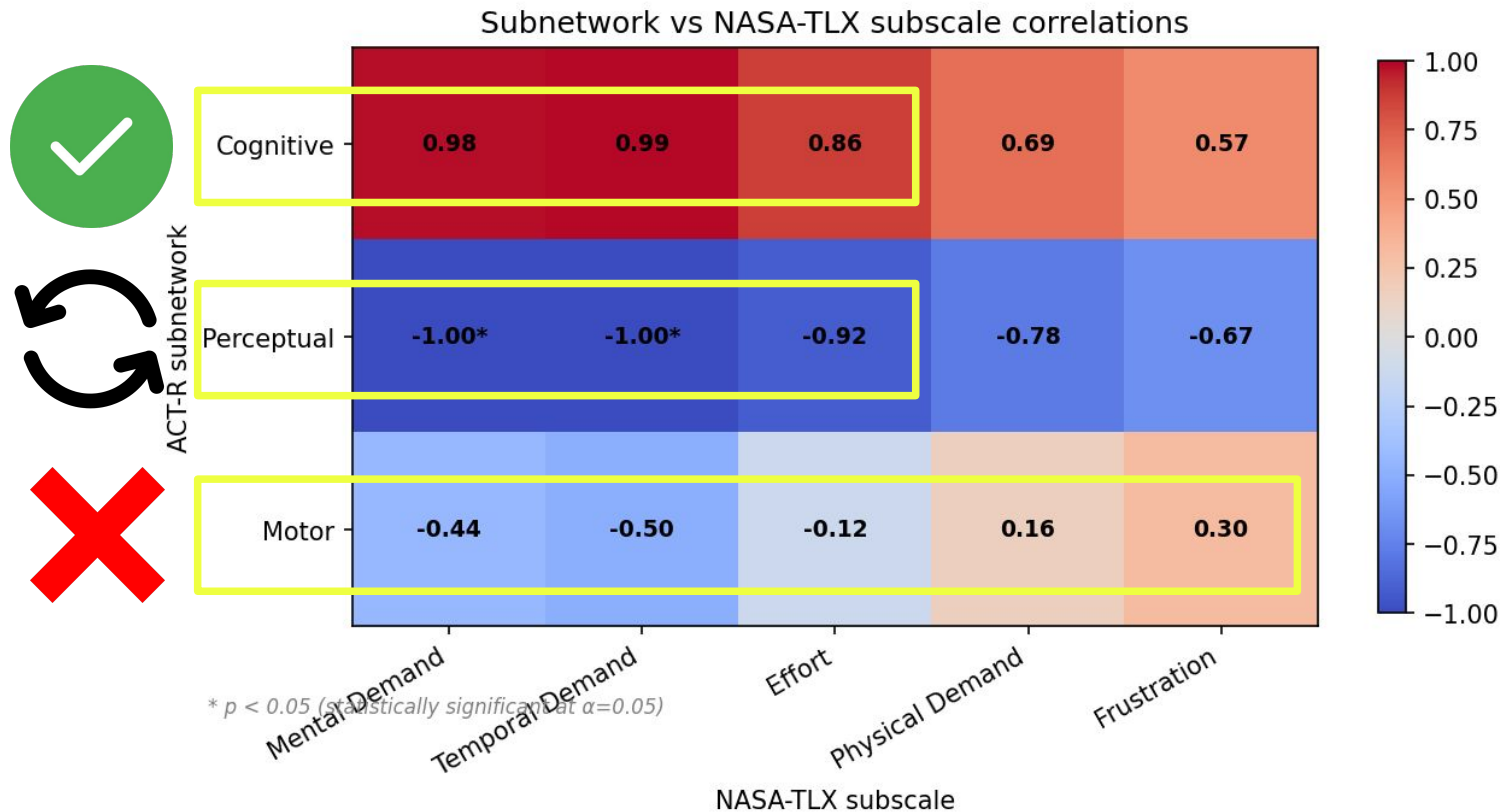
Procedure Duration: Model vs. Human Participants

BEFORE TAKEOFF

LINE-UP AND HOLD



6 FUTURE MODEL VALIDATION



Conclusion



We presented a simulation framework for exploring human–autonomy teaming design space. So far, the cognitive model has been developed for 2/5 procedures of the chosen flight scenario.

Next step: scale to the full scenario and validate against multi-level metrics (performance, interaction, cognition). Run simulations over pilot-configured allocations to cover a broader design space.

Questions? You can email me @ benjaminberton64@gmail.com



code and resources
on my github



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